

# Neural Control and Coordination

1. Given below are two statements :

**Statement-I:** Chemical synapse shows bidirectional flow.

**Statement-II:** Electrical synapses are abundant in our system.

In the light of the above statements, choose the most appropriate answer from option given below

- (1) Statement I and Statement II both are correct.
- (2) Statement I is correct, but statement II is incorrect.
- (3) Statement I is incorrect, but Statement II is correct.
- (4) Statement I and Statement II both are incorrect.

2. The neurons in all animals are capable

- i. detecting various stimuli in environment
- ii. receiving and transmitting stimuli to CNS
- iii. processing various internal and external stimuli

Select the most appropriate option.

- (1) I, II and III are correct
- (2) Only II is correct
- (3) II and III are correct
- (4) I and II are correct

3. Excitatory neurotransmitter is;

- (1) Norepinephrine
- (2) Dopamine
- (3) GABA
- (4) Serotonin

4. Which one of the following types of neurons is most numerous in the body?

- (1) Unipolar
- (2) Multipolar
- (3) Bipolar
- (4) Pseudo-unipolar

5. Match List-I with List-II

**List-I**

**List-II**

- |                       |                       |
|-----------------------|-----------------------|
| (1) Multipolar neuron | (I) Embryonic stages  |
| (2) Bipolar neuron    | (II) Retina of eyes   |
| (3) Unipolar neuron   | (III) Cerebral cortex |

Choose the most appropriate answer

- (1) a-3, b-2, c-1
- (2) a-2, b-1, c-3
- (3) a-1, b-3, c-2
- (4) a-2, b-3, c-1

6. Visceral nervous system within human body

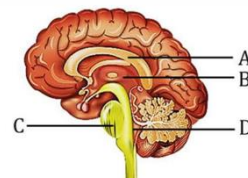
- I. is division of peripheral nervous system
- II. is the division of central nervous system
- III. consist of nerve fibres and ganglia
- IV. carry impulses from one visceral organ to another.

Which of the following option is correct

- (1) I and II are correct
- (2) II and IV are correct
- (3) I, III and IV are correct
- (4) II, III, IV are correct

7. Refer to the diagram showing sagittal section of human brain.

Identify the parts labelled as A -D.



- (1) A-Cerebrum; B-Medulla; C-Cerebellum; D-Medulla
- (2) A-Hypothalamus; B-Cerebellum; C Medulla; D-Pons

(3) A-Corpus callosum; B-Thalamus; C-Pons; D-Cerebral aqueduct

(4) A-Thalamus; B-Corpus callosum; C Medulla; D-Cerebellum

**8. During recovery, a nerve fibre becomes;**

(1) +vely charged on outside and –vely charged on inside.

(2) +vely charged on both outside and inside.

(3) –vely charged on outside and +vely charged on inside.

(4) –vely charged on both outside and inside.

**9. Select the incorrect statement regarding synapses.**

(1) Electrical current can flow directly from one neuron into the other across the electrical synapse.

(2) Chemical synapses use neurotransmitters.

(3) Impulse transmission across a electrical synapse is always slower than that across an chemical synapse.

(4) The membranes of presynaptic and postsynaptic neurons are in close proximity in an electrical synapse.

**10. Given below are two statements :**

**Assertion(A): Brain and spinal cord are protected from physical impact and trauma.**

**Reason(R): CSF acts as a cushion, protecting the brain and spinal cord.**

**Choose the most appropriate answer from following options**

(1) Both Assertion (A) and Reason (R) are true, and Reason (R) is a correct explanation of Assertion (A).

(2) Both Assertion (A) and Reason (R) are true, but Reason (R) is not a correct explanation of Assertion (A).

(3) Assertion (A) is true, and Reason (R) is false.

(4) Assertion (A) is false, and Reason (R) is true.

**11. Which of the following is not the function of hypothalamus?**

(1) Control body temperature

(2) Synthesis of hormones which are released by neurohypophysis

(3) Storage of memory and seat of intelligence

(4) Control the secretion and inhibition of hormones of the adenohypophysis

**12. The central nervous system (CNS) originates from the embryonic \_\_\_\_ (A) \_\_\_\_ which develops from the \_\_\_\_ (B) \_\_\_\_ during early stages of fetal development.**

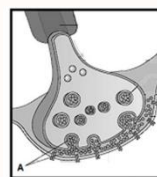
(1) A -neural tube; B – endoderm

(2) A -cartilage; B – endoderm

(3) A -neural tube; B – mesoderm

(4) A -neural tube; B – ectoderm

**13. Observe the given diagram and select the correct option regarding labelled part A.**



(1) It may be either excitatory or inhibitory

(2) It is synthesized by mitochondria

(3) It is stored in synaptic vesicles of synaptic knob

(4) All of these

**14. Consider the following statements**

**Assertion-**The speed of nerve impulse along axon would slowdown in the absence of Schwann cells  
**Reason-**Schwann cells are abundantly found in cells body of neurons.

**Choose the most appropriate answer**

(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion

(2) Both Assertion and Reason are true, but Reason is not the correct explanation of assertion

(3) Assertion is true, but Reason is false

(4) Assertion is false, but Reason is true

**15. Arrange the arachnoid, pia mater, and dura mater according to their location from inner to outer layer.**

- (1) Dura mater → Pia mater → Arachnoid
- (2) Arachnoid → Dura mater → Pia mater
- (3) Pia mater → Arachnoid → Dura mater
- (4) Pia mater → Dura mater → Arachnoid

**16. Consider the following statements.**

**(A) At rest, the axoplasm inside the axon contain low concentration of  $K^+$  ions.**

**(B) The concentration gradient across axonal membrane is generated due to the different concentration of  $Na^+$  and  $K^+$  ions across it.**

**Select the correct option.**

- (1) A is true, B is false
- (2) Both A and B are false
- (3) A is false, B is true
- (4) Both A and B are true

**17. Match List-I with List-II**

| List-I        | List-II                          |
|---------------|----------------------------------|
| I. Gyrus      | A. A very thin middle layer      |
| II. Arachnoid | B. Ridges                        |
| III. Sulci    | C. Grooves                       |
| IV. Pia mater | D. Contact with the brain tissue |

**Select the correct option.**

- (1) I-A, II-B, III-D, IV-C
- (2) I-C, II-D, III-B, IV-A
- (3) I-B, II-A, III-C, IV-D
- (4) I-C, II-D, III-A, IV-B

**18. Semi circular canals are :-**

- (1) Three in number and lies at right angle to each other
- (2) Three in number & present in cochlea near helicotrema.
- (3) Present in middle ear
- (4) Four in number and lies at right angle to each other.

**19. Given below are two statements :**

**Assertion(A): The inner portion of the cerebellum is composed of grey matter.**

**Reason(R): Excessive alcohol consumption can lead to cerebellar atrophy, impacting the cerebellum's structure and function.**

**In the light of the above statements, choose the most appropriate answer from option given below**

- (1) Both Assertion (A) and Reason (R) are true, and Reason (R) is a correct explanation of Assertion (A).
- (2) Both Assertion (A) and Reason (R) are true, but Reason (R) is not a correct explanation of Assertion (A).
- (3) Assertion (A) is true, and Reason (R) is false.
- (4) Assertion (A) is false, and Reason (R) is true.

**20. Which of the following statement(s) is/are correct about the cortex of cerebrum?**

**I. It consists of grey matter.**

**II. It shows prominent folds.**

**III. It consists of white matter.**

**IV. It contains motor areas, sensory areas and association areas.**

**Choose the correct option from the options given below.**

- (1) Only I
- (2) I and II
- (3) I, II and IV
- (4) I, III and IV

**21. Resting membrane potential across the nerve cell is maintained by**

- (1) Opening of voltage-gated  $Na^+$  channels
- (2) Opening of voltage-gated  $K^+$  channels
- (3) Action of  $Na^+ - K^+$  ATPase pump
- (4) Influx of  $Na^+$  from ECF

**22. Match the column-I with column-II**

|      | Column-I           |    | Column-II                                                           |
|------|--------------------|----|---------------------------------------------------------------------|
| I.   | Brain stem         | A. | Auditory processing                                                 |
| II.  | Superior colliculi | B. | Control respiration, cardiovascular reflexes and gastric secretions |
| III. | Inferior colliculi | C. | Mid brain, pons and medulla oblongata                               |
| IV.  | Medulla oblongata  | D. | Visual processing                                                   |

- (1) I-A, II-B, III-D, IV-C
- (2) I-C, II-D, III-B, IV-A
- (3) I-B, II-A, III-C, IV-D
- (4) I-C, II-D, III-A, IV-B

**23. In humans, the brain stem connects**

- (1) Medulla oblongata and pons only
- (2) Midbrain and cerebellum only
- (3) Two cerebellar hemispheres
- (4) Brain with spinal cord

**24. Arrange the following cranial meninges in correct sequence from outer to inner side.**

- A – Pia mater
- B – Arachnoid
- C – Dura mater

**Choose the correct option**

- (1) A – B – C
- (2) B – A – C
- (3) C – B – A
- (4) B – C – A

**25. In humans, the part of the brain which receives and integrates visual, tactile and auditory inputs is**

- (1) Forebrain
- (2) Hindbrain
- (3) Hypothalamus
- (4) Midbrain

**26. Repolarisation of a nerve fibre is characterised by the**

- (1) Rapid influx of K<sup>+</sup>
- (2) Rapid efflux of K<sup>+</sup>
- (3) Rapid efflux of Ca<sup>2+</sup>
- (4) Rapid influx of Na<sup>+</sup>

**27. Which of the following types of neurons possess only one axon?**

- a. Multipolar neurons
- b. Bipolar neurons
- c. Unipolar neurons

**Choose the correct option from the options given below.**

- (1) Only a and b
- (2) Only b and c
- (3) Only a and c
- (4) a, b and c

**28. In the visceral neural system, the impulses travel**

- (1) Only from CNS to viscera
- (2) Only from viscera to CNS
- (3) From CNS to viscera and vice versa
- (4) From CNS to skeletal muscles

**29. If the sodium-potassium pump of a neuron fails after depolarisation, what immediate effect would this likely have on neuron's function?**

- (1) Voltage gated sodium channels will remain open for longer duration
- (2) The neuron will not conduct another action potential until the resting potential is fully restored
- (3) The concentration of sodium ions and potassium ions would be equal in both outside the cell and inside the cell
- (4) The permeability of neuronal membrane to chloride ions would increase dramatically



**30. Read the following statements w.r.t PNS present in human and choose the correct option.**

**Statement A: The myelinated nerve fibres are enveloped with Schwann cells and are found in spinal nerves.**

**Statement B: Schwann cells do not enclose the unmyelinated nerve fibres.**

- (1) Only statement A is correct
- (2) Only statement B is correct
- (3) Both statements A and B are correct
- (4) Both statements A and B are incorrect

**31. The structure in the neuron which receives impulses from different sources and directs them towards the cell body is**

- (1) Axon
- (2) Synaptic knob
- (3) Dendrite
- (4) Myelin sheath

**32. Which pair of structures are exclusively present in nerve cells?**

- (1) Nucleus and mitochondria
- (2) Cyton and dendrites
- (3) Vacuoles and fibres
- (4) Flagellum and medullary sheath

**33. Assertion (A) : There is delay in the transmission of nerve impulses across chemical synapses.**

**Reason (R) : Synaptic delay results from the time taken in releasing the neurotransmitter and in stimulating the post synaptic neuron.**

**In the light of above statements, select correct option.**

- (1) Both (A) and (R) are true and (R) is the correct explanation of the (A)
- (2) Both (A) and (R) are true but (R) is not the correct explanation of the (A)
- (3) (A) is true statement but (R) is false
- (4) Both (A) and (R) are false statements

**34. Choose the incorrect match w.r.t. the part of brain and its functions**

- (1) Cerebellum : Balance and coordinating locomotory activities of the body
- (2) Association areas : Memory and communication
- (3) Limbic system : Regulation of sexual behaviour
- (4) Medulla oblongata : Urge for eating and drinking

**35. Sympathetic neural system can be differentiated from parasympathetic neural system in all of the following aspects, except**

- (1) Former increases the strength of ventricular contractions
- (2) Latter is a part of the autonomic nervous system
- (3) Former increases the heart rate
- (4) Former increases the cardiac output

**36. Ventral root ganglion;**

- (1) sends motor signals from the peripheral nervous system to the central nervous system.
- (2) sends sensory signals from the central nervous system to the peripheral nervous system.
- (3) sends motor signals from the central nervous system to the peripheral nervous system.
- (4) sends sensory signals from the peripheral nervous system to the central nervous system.

**37. Which of the following is/are not regulated by limbic system?**

- (i) Sexual behaviour
- (ii) Excitement
- (iii) Body balance
- (iv) Communication

- (1) Only ii
- (2) ii and iii
- (3) iii and iv
- (4) ii and iv

**38. Arrange the arachnoid, pia mater, and dura mater according to their location from inner to outer layer.**

- (1) Dura mater → Pia mater → Arachnoid
- (2) Arachnoid → Dura mater → Pia mater
- (3) Pia mater → Arachnoid → Dura mater
- (4) Pia mater → Dura mater → Arachnoid

**39. Select the incorrect statement regarding synapses.**

- (1) Electrical current can flow directly from one neuron into the other across the electrical synapse.
- (2) Chemical synapses use neurotransmitters.
- (3) Impulse transmission across a electrical synapse is always slower than that across an chemical synapse.
- (4) The membranes of presynaptic and postsynaptic neurons are in close proximity in an electrical synapse.

**40. Select incorrect statements.**

- (i) The electrical potential difference across the resting plasma membrane is called as the action potential.
- (ii) During polarisation, the outer surface of the axonal membrane possesses a positive charge while its inner surface becomes negatively charged.
- (iii) During resting, the membrane is permeable to negatively charged proteins present in the axoplasm.

**Choose the correct option from the options given below.**

- (1) i and iii
- (2) i and ii
- (3) ii and iii
- (4) None of these

**41. Neural stimulation in visceral organs in human being is done by :**

- (1) Sympathetic and parasympathetic nerves and is under involuntary action
- (2) Sympathetic nerves and is under voluntary action
- (3) Sympathetic and Parasympathetic nerves and is under voluntary action
- (4) Parasympathetic nerves and is under voluntary action

**42. The third ventricle of the brain is situated in the :**

- (1) Base of telencephalon
- (2) Roof of metencephalon
- (3) Roof of diencephalon
- (4) Base of myelencephalon

**43. Read the following statements carefully and select the correct option :**

**I. The medulla is connected to the spinal cord**

**II. Medulla contains controlling centres for respiration, cardiovascular reflexes and gastric secretion.**

**III. Cerebellum have convoluted surface in order to provide the additional space for more neurons.**

- (1) Only I
- (2) I and II
- (3) Only III
- (4) I, II and III

**44. Which one of the following statements is true :**

- (1) Saltatory conduction is seen in non-myelinated nerve fibres
- (2) Nissl's granules are found in muscle fibres
- (3) Non-myelinated nerve fibres do not possess nodes of ranvier
- (4) Non- myelinated nerve fibres are completely enclosed by myelin sheath

45. Which category of nervous system comprises the whole complex of nerves, fibres, ganglia and plexus by which impulses travel from central nervous system to involuntary organs and vice versa?

- (1) Peripheral nervous system
- (2) Sympathetic nervous system
- (3) Somatic nervous system
- (4) Visceral nervous system

46. Choose the correct option from codes given below.

I. More than half volume of the neural tissue is formed by neuroglia cells

II. Astrocytes play a role in the formation of the blood brain barrier.

III. Microglia engulf microbes and cellular debris

IV. Astrocytes, oligodendrocytes and microglia, are three different types of neuroglial cells.

- (1) I and IV are correct only
- (2) II and IV are correct only
- (3) All are incorrect
- (4) All are correct

47. Match the column-I with column-II and choose the correct combination from the options given :

| Column-I           | Column-II      |
|--------------------|----------------|
| (A) Diencephalon   | (1) Cerebellum |
| (B) Telencephalon  | (2) Medulla    |
| (C) Myelencephalon | (3) Amygdala   |
| (D) Metencephalon  | (4) Thalamus   |

- (1) A - 4, B - 3, C - 1, D - 2
- (2) A - 3, B - 4, C - 1, D - 2
- (3) A - 4, B - 3, C - 2, D - 1
- (4) A - 4, B - 1, C - 2, D - 3

48. The thermoregulatory centre is situated in :

- (1) Spinal cord
- (2) Pituitary body
- (3) Cerebellum
- (4) Hypothalamus

49. Four healthy people in their twenties get involved in injuries resulting in damage which of the cells are least likely to be replaced by new cells :

- (1) Osteocytes
- (2) Malpighian layer of the skin
- (3) Liver cells
- (4) Nerve cells

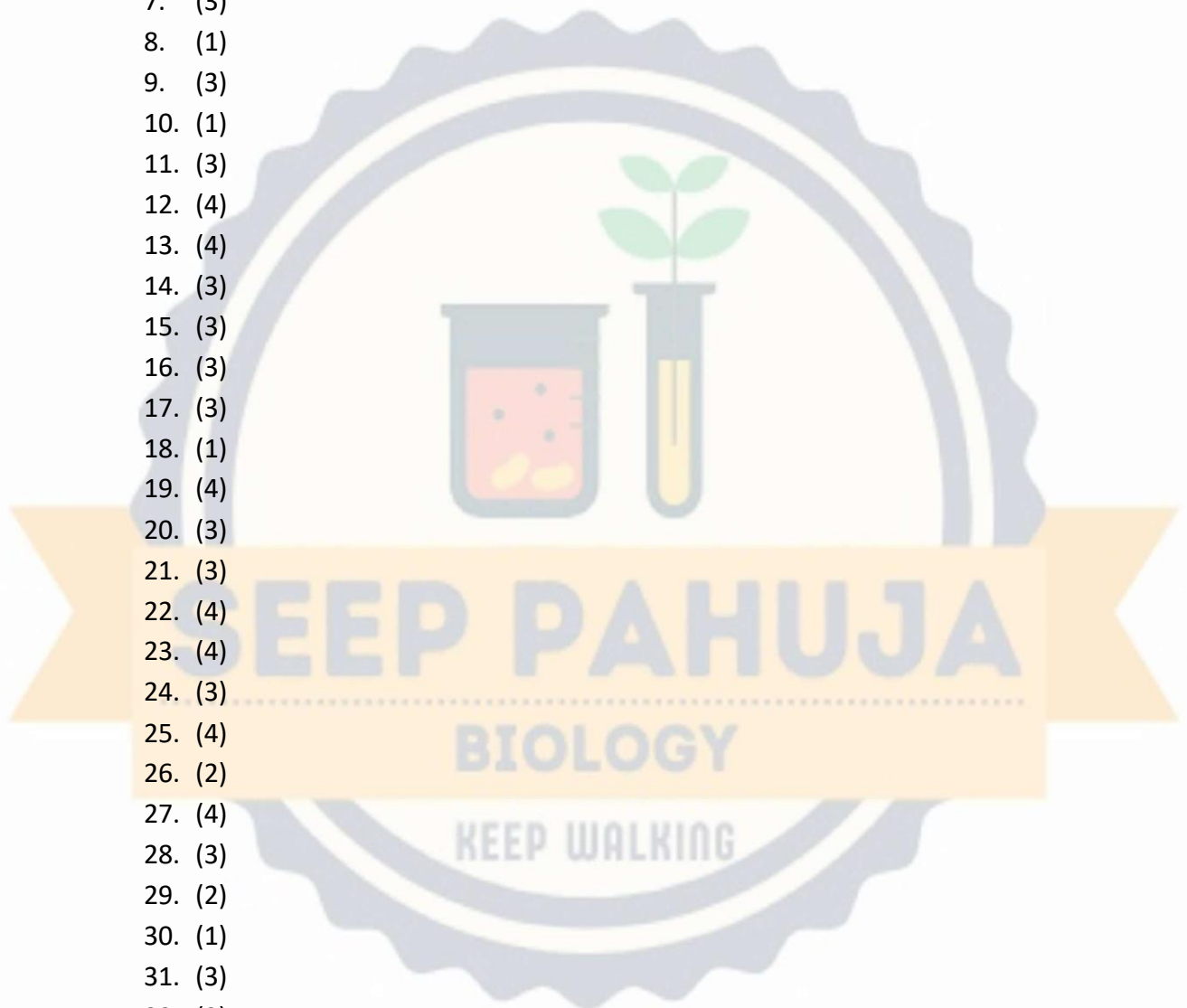
50. Assertion : Transmission of the nerve impulse across a chemical synapse is accomplished by neurotransmitters.

Reason : Transmission across a synapse usually requires neurotransmitters because there is small space i.e, synaptic cleft, that separates one neuron from another.

- (1) Both Assertion & Reason are true and the Reason is the correct explanation of the Assertion
- (2) Both Assertion & Reason are true but the Reason is not the correct explanation of the Assertion
- (3) Assertion is true statement but Reason is false
- (4) Both Assertion and Reason are False statements

# Answer Key

1. (4)
2. (1)
3. (1)
4. (2)
5. (1)
6. (1)
7. (3)
8. (1)
9. (3)
10. (1)
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40. (1)
41. (1)





- 42. (3)
- 43. (4)
- 44. (3)
- 45. (4)
- 46. (4)
- 47. (3)
- 48. (4)
- 49. (4)
- 50. (1)

